SERVICENOW SCRIPT EXECUTION FLOW

1 BEFORE FORM IS LOADED

Display Business Rule (Forms)

- Runs on the server before the form is requested.
- Use g_scratchpad to pass data to the client.

Data sent to client using g_scratchpad

2 FORM LOADING

OnLoad Client Script

- Executes when the form starts loading on the client.

UI Policies

- UI Policies are evaluated after OnLoad Client Script.
- Controls field behavior (mandatory, read-only, visible, etc.).

Form is being prepared for the user.

3 FORM LOADING

OnChange Client Script

- Executes when a field salue is changed by the user.

UI Policies

- UI Policies are evaluated after OnChange Client Script.
- Controls field behavior.

User can interact with the form.

4 FORM SUBMITTED (Submit/Update Clicked)

OnChange Client Script

- Executes when the user clicke changed/Update button.

UI Policies / Client-side Validations

- UI Policies run and can block submission if:
  - Mandatory fields missing
  - Invalid values
  - Read-only fields have data

If any validation fails, submission is blocked.

All validations passed?

- No → Show error and stop submission
- Yes →

5 BEFORE DATA INSERTED (Before Insert/Update)

Data Policy (Client + Server)

- Validates mandatory fields, read-only fields, data types, reference checks, etc.

Before Business Rule

- Runs on the server before data is written to the database.
- Perform validations, duplicacy checks, data manipulation, etc.

If all checks pass, data is ready to be inserted/ updated in the database.

6 AFTER DATA INSERTED (After Insert/Update)

After Business Rule (Same Transaction)

- Runs immediately after the record is inserted/updated in the database.
- Use for real-time actions that must complete instantly.

OR

Async Business Rule (Background)

- Runs asynchronously after the transaction is committed.
- Recommended for non-urgent operations (email, integrations, notifications, heavy processing, etc.).

Post-processing actions can be performed.

Flow repeats for the next request

EXECUTION ORDER SUMMARY

1 Before Form Is Loaded	2 Form Loading	3 Form Loaded	4 Form Submitted	5 Before Data Inserted	6 After Data Inserted (After / Async)
Display BR runs on the server and sends data to client using g_scratchpad.	OnLoad Client Script runs → UI Policies run → Form is prepared.	OnChange Client Script runs on field change → UI Policies run again.	OnSubmit Client Script runs → UI Policies & validations run → submission allowed.	Data Policy validates → Before Business Rule runs → Data ready for DB.	After Business Rule runs immediately OR Async Business Rule runs in background.

AFTER VS ASYNC BUSINESS RULES

AFTER BUSINESS RULE

- Runs immediately after the record is inserted/updated.
- Executes in the same transaction.
- User waits until it completes.
- Use for critical, real-time updates.

Example:
Update related record, recalculate fields, send immediate response to user.

VS

ASYNC BUSINESS RULE

- Runs asynchronously after the transaction is committed.
- Executes in the background (separate thread/queue).
- User does NOT wait.
- Use for heavy processing, emails, notifications, integrations, etc.

Example:
Send email notification, create multiple records, call external API, generate reports.

KEY TAKEAWAYS

- Display BR + g_scratchpad helps pass data to the client efficiently.
- UI Policies control the form behavior at every stage.
- Use Client Scripts for user interactions (OnLoad, OnChange, OnSubmit).
- Always let Data Policy and Before BR validate the data before it goes to the database.
- Use After BR only when immediate action is required.
- Prefer Async BR for better performance and user experience.

SERVICENOW SCRIPT EXECUTION FLOW

1 Before form is loaded

Display Business Rule (Forms)

- Runs on the server before the form is requested.
- Use g_scratchpad to pass data to the client.

Data sent to client using g_scratchpad

2 Form loading

OnLoad Client Script

- Executes when the form starts loading on the client.

UI Policies

- UI Policies are evaluated after OnLoad Client Script.

Form is being prepared for the user.

3 Form loaded

OnChange Client Script

- Executes when a field value is changed by the user.

UI Policies

- UI Policies are evaluated after OnChange Client Script.

User can interact with the form.

4 Form submitted (Submit/Update clicked)

OnSubmit Client Script

- Executes when the user clicks Submit/Update button.

UI Policies / Client-side Validations

- UI Policies run and can block submission if:
  - Mandatory fields missing
  - Invalid values
  - Read-only fields have data

If any validation fails, submission is blocked.

All validations passed?

- No → Show error and stop submission
- Yes → 5

5 Before data inserted (Before Insert/Update)

Data Policy (Client + Server)

- Validates mandatory fields, read-only fields, data types, reference checks, etc.

Before Business Rules

- Runs on the server before data is written to the database.
- Perform validations, duplicacy checks, data manipulation, etc.

If all checks pass, data is ready to be inserted/ updated in the database.

6 After data inserted (After Insert/Update)

After Business Rules

- Runs after the record is inserted/updated in the database.
- Use for synchronous operations that need inmediate execution.

OR

Async Business Rules (AC)

- Runs asynchronously after the transaction is committed.
- Recommended for non-urgent operations (e.g., integrations, notifications, heavy processing).

Post-processing actions can be performed.

Flow repeats for the next request

	1 Before form is loaded	2 Form loading	3 Form loaded	4 Form submitted	5 Before data inserted	6 After data inserted
STAGE OVERVIEW	Display Business Rules run on the server before the form is sent to the client. Use g_scratchpad to pass data to the client.	OnLoad Client Script executes when the form starts loading. Then UI Policies run to control field behaviors.	OnChange Client Script runs when a field value changes. UI Policies run again after OnChange.	OnSubmit Client Script runs when Submit/Update is clicked. UI Policies and client validations run and can block submission.	Data Policy validates the data. Before Business Rules run on the server before data is written to the database.	After Business Rules run synchronously after the data is inserted/updated. Async (AC) Business Rules run asynchronously.
KEY POINTS	• Runs once per form load • Cannot access g_form • Use g_scratchpad to pass data to client	• OnLoad runs once • UI Policies evaluate after OnLoad • Good for initial setup	• OnChange runs on each field change • UI Policies run after OnChange	• OnSubmit runs on submit • UI Policies can block submission with errors • All client checks happen here	• Data Policy enforces data integrity • Before BR can block insert/ update by setting error	• After BR runs in same transaction • Async BR runs in background (recommended)

BEST PRACTICES

- Use Display BR + g_scratchpad to pass data to the client.
- Use UI Policies for UI behavior instead of Client Scripts whenever possible.
- Keep OnChange and OnSubmit scripts lightweight.
- Use Before Business Rules for validations and data manipulation.
- Use Async Business Rules for integrations, notifications, and heavy processing.